

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T45 · Volume and Surface Area

Questions

Student book · questions with writing space

■ 19 pages

FAMILY MAP

Volume Surface Area

F01	Find volume of prisms and cylinders	7 Q 30 marks
F02	Find surface area of prisms and cylinders	4 Q 15 marks
F03	Resolve composite solids and volume-unit constraints	5 Q 23 marks
F04	Use sphere and cone formulae	2 Q 9 marks

7 Here is a solid cylinder.

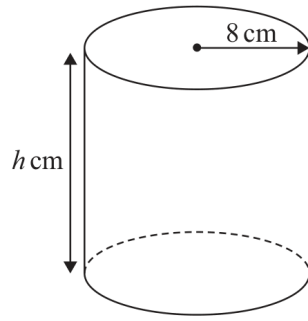


Diagram **NOT**
accurately drawn

The radius of the cylinder is 8 cm

The height of the cylinder is h cm

The volume of the cylinder is 3892 cm^3

Work out the value of h

Give your answer correct to one decimal place.

$h = \dots\dots\dots$

(Total for Question 7 is 3 marks)

- 21 The diagram shows a cuboid.

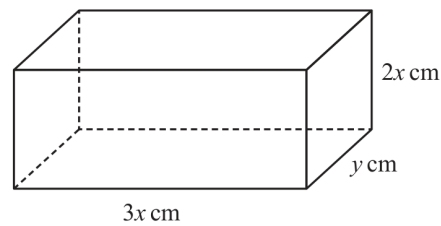


Diagram **NOT**
accurately drawn

The cuboid measures $3x$ cm by $2x$ cm by y cm

The volume of the cuboid is 1014 cm^3

The total surface area of the cuboid is $A\text{ cm}^2$

Show that $A = 12x^2 + \frac{1690}{x}$

You must show all the stages of your working.

(Total for Question 21 is 3 marks)

Question 10

4MA1-1H Q23

MODULAR UNIT 2

5 marks

- 23 A solid shape is made by removing a hemisphere, shown shaded, from a cone as shown in the diagram.

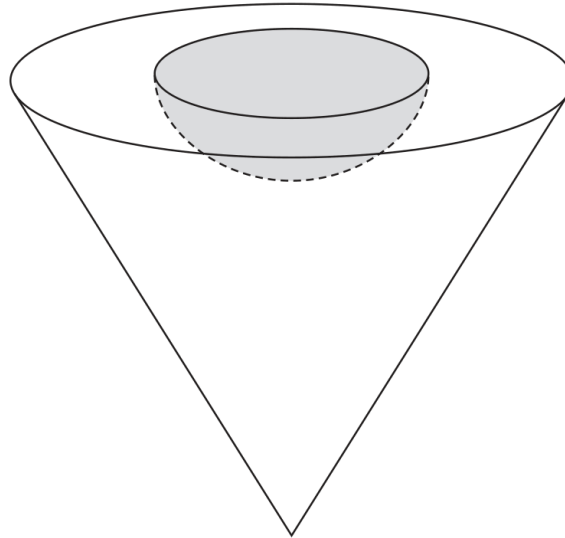


Diagram **NOT**
accurately drawn

The radius of the hemisphere is $2x$ cm

The radius of the base of the cone is $5x$ cm

The vertical height of the cone is $6x$ cm

The volume of the solid shape is 6948π cm³

Work out the **total** surface area of the solid hemisphere that has been removed from the cone.

Give your answer correct to the nearest integer.

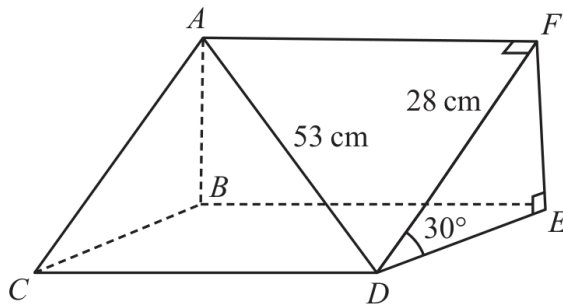
..... cm²

Question 16

4MA1-1H Q21 demand

MODULAR UNIT 2

5 marks

21 Here is a triangular prism $ABCDEF$ Diagram **NOT**
accurately drawn

$$AD = 53 \text{ cm}$$

$$DF = 28 \text{ cm}$$

$$\text{Angle } FDE = 30^\circ$$

Work out the volume of the triangular prism.

Give your answer correct to the nearest whole number.

..... cm^3

Question 04

4MA1-1H Q5 M01

MODULAR UNIT 2

4 marks

- 5 The diagram shows a cylinder.

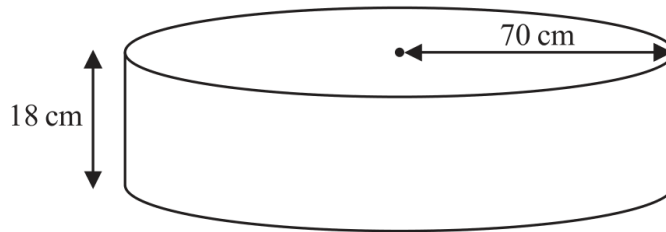


Diagram **NOT**
accurately drawn

The radius of the cylinder is 70 cm

The height of the cylinder is 18 cm

Work out the volume of the cylinder.

Give your answer in litres correct to the nearest litre.

..... litres

Question 14

4MA1-1H Q22 M01

MODULAR UNIT 2

3 marks

- 22 The diagram shows a bowl in the shape of a hemisphere.
The bowl is made from metal.

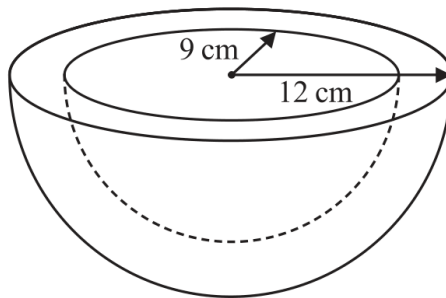


Diagram **NOT**
accurately drawn

The outer radius of the bowl is 12 cm
The inner radius of the bowl is 9 cm

The thickness of the bowl is uniform.

Work out the volume of the metal.
Give your answer correct to the nearest whole number.

..... cm³

Question 20

4MA1-2H Q6 M01

MODULAR UNIT 2

3 marks

- 6 The diagram shows a solid triangular prism.

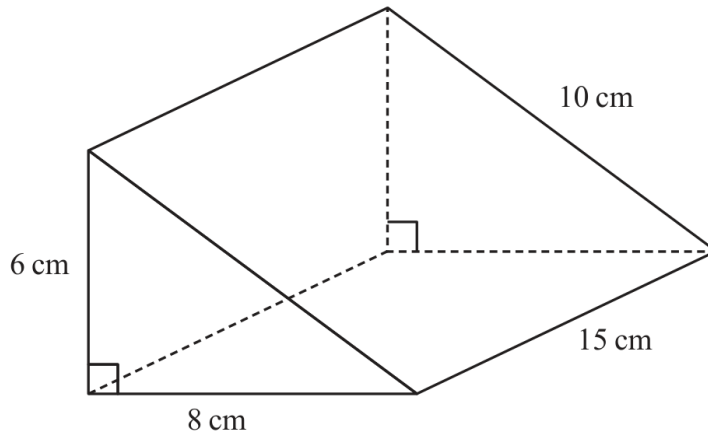


Diagram **NOT**
accurately drawn

Work out the **total** surface area of the triangular prism.

..... cm^2

Question 08

4MA1-1H Q7 Q7

MODULAR UNIT 2

4 marks

- 7 A cylinder is placed on the ground.

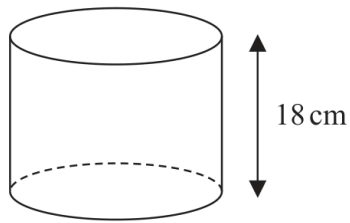


Diagram **NOT**
accurately drawn

The height of the cylinder is 18 cm.

The force exerted by the cylinder on the ground is 72 newtons.

The pressure on the ground due to the cylinder is 1.4 newtons/cm²

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the volume of the cylinder.

Give your answer correct to 3 significant figures.

..... cm³

Question 18

4MA1-1H Q22 Q22

MODULAR UNIT 2

5 marks

- 22 A solid is made from a cone and a hemisphere.

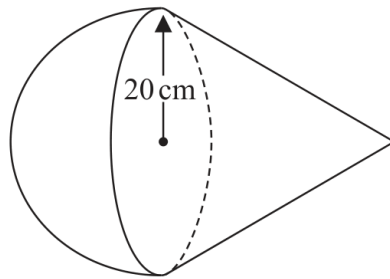


Diagram **NOT**
accurately drawn

The circular plane face of the hemisphere coincides with the circular base of the cone.
The radius of the hemisphere and the radius of the circular base of the cone are both 20 cm.

The curved surface area of the cone is $580\pi\text{cm}^2$

The volume of the solid is $k\pi\text{cm}^3$

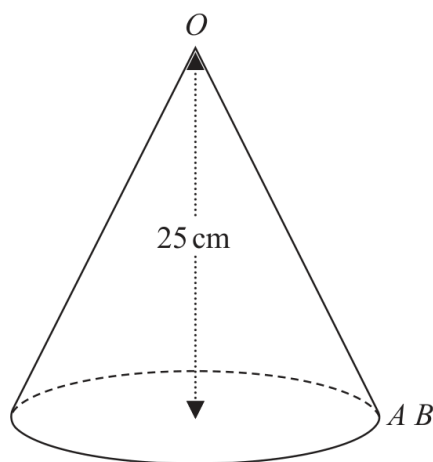
Work out the exact value of k

$k = \dots\dots\dots$

-
- A 3D diagram of a cube. The three edges that are visible (the front-bottom edge, the front-left vertical edge, and the receding-right edge) are each labeled with the text $w\text{ cm}$.

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$
..... cm³

The sector can form the curved surface of a cone by joining OA to OB .



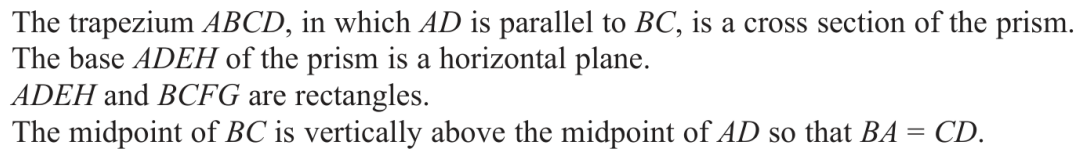
The height of the cone is 25 cm.
The volume of the cone is 1600 cm^3

Work out the value of x .
Give your answer correct to the nearest whole number.

- Diagram **NOT**
accurately drawn



Diagram **NOT**
accurately drawn



The perpendicular distance between edges AD and BC is 20 cm.

$$\dots\dots\dots \text{cm}^2$$

A 3D diagram of a triangular prism. The base is a right-angled triangle with legs of 7.4 cm and 15 cm, and a hypotenuse of 11.2 cm. The height of the prism is indicated by a dashed line perpendicular to the base.

Diagram **NOT**
accurately drawn

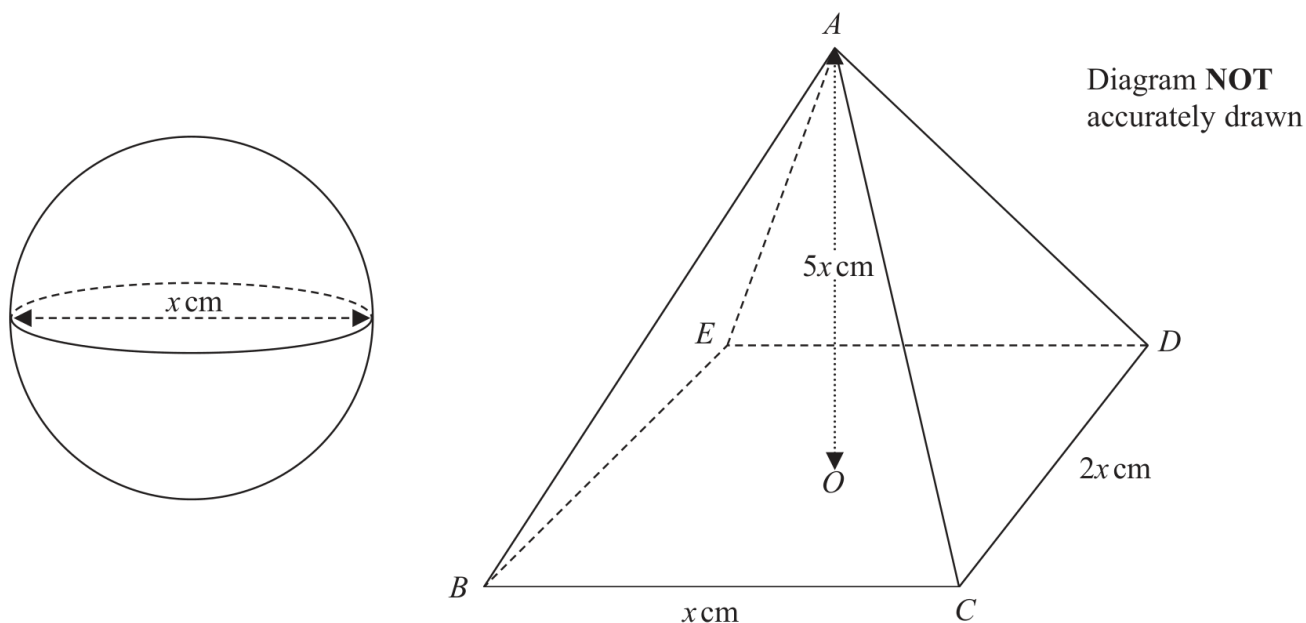
..... cm³

A diagram of a cylinder. The top circular face has a radius indicated by a horizontal line from the center to the edge, labeled "10 cm". The height of the cylinder is indicated by a vertical line on the right side, labeled "h cm". The bottom circular face is represented by a dashed line to show it is hidden from view.

The mass of the cylinder is 5.4 kg.
The density of aluminium is 0.0027 kg/cm³

Calculate the value of h .
Give your answer correct to one decimal place.

- 22 The diagram shows a sphere of diameter x cm and a pyramid $ABCDE$ with a horizontal rectangular base $BCDE$.



The vertex A of the pyramid is vertically above the centre O of the base so that $AB = AC = AD = AE$.

$BC = x$ cm, $CD = 2x$ cm and $AO = 5x$ cm.

The volume of the sphere is 288π cm³

Calculate the total surface area of the pyramid.

Give your answer correct to the nearest cm²

..... cm²

- 9 The diagram shows a solid cylinder with radius 3 m.

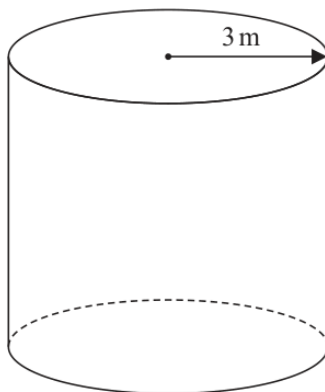


Diagram **NOT**
accurately drawn

The volume of the cylinder is $72\pi \text{ m}^3$

Calculate the **total** surface area of the cylinder.
Give your answer correct to 3 significant figures.

..... m^2

Give your answer correct to the nearest integer.

.....cm³

(Total for Question 21 is 3 marks)